IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

EEE PART III/IV: MEng, BEng and ACGI

ELECTRICAL ENERGY SYSTEMS

Wednesday, 13 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer 2 questions from Section A and 2 questions from Section B.

Use a separate answer book for each section.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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ELEC60012 – Electrical Energy Systems

Formula sheet

Part A

1. Coincidence coefficient for *n* households

$$j_n = j_\infty + \frac{1 - j_\infty}{\sqrt{n}}$$
$$P_n = j_n \times n \times P_1$$

2. Gauss-Seidel power flow update law

$$PQ\ bus: V_i^k = \frac{1}{Y_{ii}}\left(\frac{S_i^*}{(V_i^{k-1})^*} - \sum_{j=1}^{i-1} Y_{ij}\, V_j^k - \sum_{j=i+1}^{Nbus} Y_{ij}\, V_j^{(k-1)}\right)$$

Part B

1. Short circuit level in MVA

$$SCL\ (\text{in MVA}) = \frac{SCL_{base}\ (\text{in MVA})}{X_{eq}\ (\text{p.u.})}$$

2. Transformation matrix for sequence domain (a-phase) to phase domain and vice versa

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

The same transformation matrices are valid for current as well

3. Swing equation

$$\left(\frac{2H}{\omega_s}\right)\frac{d^2\delta}{dt^2} = P_m - P_e\ \text{p.u.}$$

Section A – Answer any 2 out of 3 questions in section A

Question 1

a) Assume that there are three different generators with capacities of 40kW, 50 kW and 60 and availability of 99%, 98% and 97% respectively.

(i) What are the probabilities of meeting a peak demand of 148 kW, 105 kW and 97 kW respectively?

[5]

(ii) Assume that the coincidence factor for an infinite number of households is 0.444. The peak demand of each individual household is 5 kW and the reliability criterion requires that the probability of meeting peak demand is above 95%, how many households can be supplied by these generators?

[5]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Dynamic Containment (DC) in the UK. DC reduces the need for the regular frequency response (R), following this rule

$$\left(\frac{H}{f_0} - \mathrm{DC} \times \frac{\mathrm{T_{DC}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_{\mathrm{L}} - \mathrm{DC})^2 \times \mathrm{T_d}}{4 \times \Delta f_{\max}}$$

Nomenclature:

H: System level of inertia (MW-s)
R: Amount of frequency response scheduled (MW)
DC: Amount of DCscheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
f_0: Nominal frequency of the power grid (Hz)
$\Delta f_{\max}$: Maximum admissible frequency deviation (Hz)
$\mathrm{T_d}$: Delivery time of frequency response (s)
T_{DC}: Delivery time of DC (s)

Assumptions:
1. The system level of inertia (H) of 150 GWs, i.e., H = 150 GWs,
2. The cost of frequency response (R): \$5/MW,
3. The largest generation unit in the UK: 1.8 GW,
4. Other numerical values: f_0 = 50 Hz, $\Delta f_{\max}$ = 0.8 Hz, $\mathrm{T_d}$ = 10 s, $\mathrm{T_{DC}}$ = 0.5 s.

(i) Explain why faster frequency reserve (such as DC) is more effective in limiting the frequency nadir.

[2]

(ii) Derive the closed-form expression of DC as a function of frequency response (R) when the frequency nadir reaches the limit. Calculate how much DC is needed to maintain the nadir constraint if the purchased R by the system operator is 3000MW, and 2000MW respectively. Sketch the relationship between DC and R within the range $R \in [0,3.375]$ GW, through a smooth curve, by evaluating different R, (e.g. $R \in \{0,0.5,1,2,3\}$ GW).

[4]

(iii) If the price of DC is 15 £/MW, determine the optimal amount of DC and R the system operator should purchase to achieve the minimum cost.

[4]

Question 2

a) What are the steps to follow when converting to a per-unit system in the analysis of power systems?

[3]

b) Consider the high-voltage transmission circuit in Figure 2-1. The current magnitude at the receiving end is 2 kA (lagging).

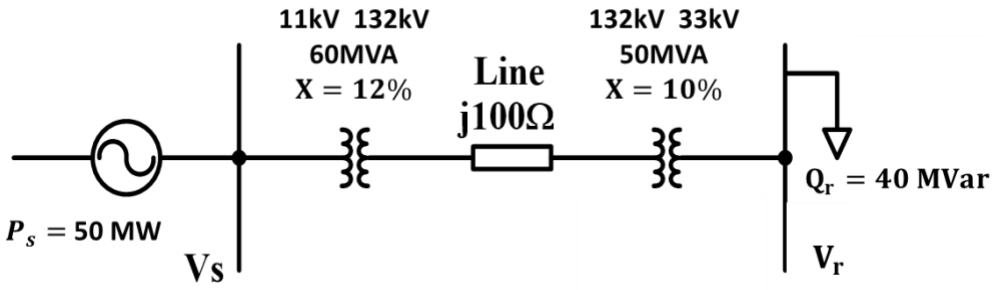


Figure 2-1 Transmission circuit with powers specified at both sending and receiving ends

(i) Calculate the per unit values of the quantities (power and impedance) in the figure, with properly selected bases.

[3]

(ii) Calculate Vs and Qs in kV and MVar respectively and compare the relationship between the sending and receiving end powers.

[5]

c) Write down the general power flow equations of a two-bus system (seen from the sending end, neglecting the shunt admittance). Derive the sensitivities of the active and reactive power with respect to the angle difference and the sending end voltage magnitude (four elements), if the line resistance is zero. What conclusions can be made in this case, regarding the PQ-Vθ coupling? Further assume that the voltage magnitudes at both ends are 1 pu and the voltage angle difference is small. Derive the DC approximation of the power flow equations and extend it to a multi-bus system.

[9]

Question 3

A three-bus power network is presented in Figure 3.1 below. Data relevant for the load flow analysis on this system are given in per-unit values.

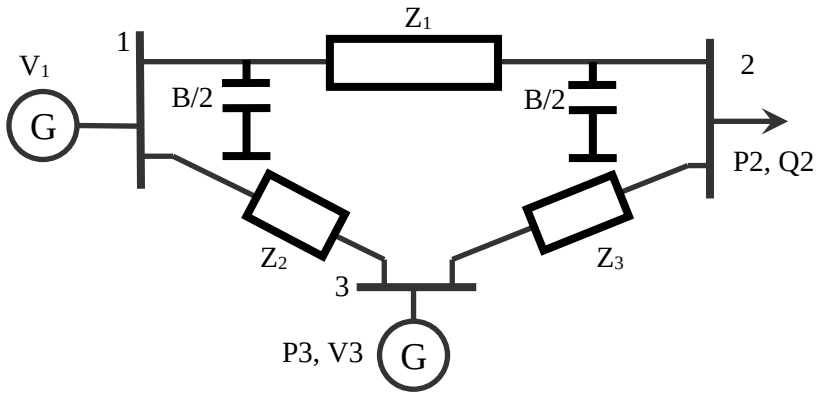


Figure 3.1 A three-bus power network

a) Express the bus admittance matrix for the system in terms of Z_1, Z_2, Z_3 and B.

[5]

b) Assuming the per unit values |V1| = 1, |V3| = 1.04, P3 = 1.02, P2 = 0.97, Q2 = 1, Z_1 = 0.25j, Z_2 = 0.25j, Z_3 = 0.2j and B = 0, perform two iterations of the Gauss-Seidel load flow method.

[12]

c) What are the consequences if the voltage magnitude at bus 2 is too low? What measures can be taken in this case?

[3]

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) State the four steps for fault calculation without making any simplifying assumptions about pre-fault voltages or currents. No mathematical expressions are necessary.

[4]

b) The strength of an AC system at a particular bus is dictated by the short circuit level. Justify this statement.

[4]

c) Explain why steady-state reactance of synchronous generators is not used for fault analysis.

[5]

d) Three 20 kV generators (G_1, G_2, G_3) are connected as shown in Figure 4.1. The parameters of the generators and the reactors are given in Table 4.1.

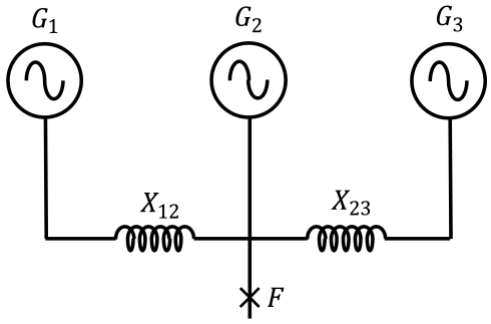


Figure 4.1: Connection of three generators and the fault location

	Rating (MVA)	X (p.u.)	X'' (p.u.)
G_1	45	0.2	0.1
G_2	50	0.3	0.15
G_3	40	0.2	0.1
X_{12}	45	0.2	-
X_{23}	40	0.3	-

Table 4.1: Parameters of generators and reactors

For a 3-phase fault at F with zero fault impedance, calculate the following using a 50 MVA base and assuming the pre-fault voltage to be 1.0 p.u.:

i) Short circuit level in MVA.

[5]

ii) Fault current in kA.

[2]

Question 5

a) For each of the following two types of unbalanced faults, derive the connection between the sequence networks starting from fault current and voltage in phase domain:

i) Line to ground (LG) fault on phase 'a' with a fault impedance Z_f.

[5]

ii) Line to line (LL) fault between phase 'b' and 'c' with a fault impedance Z_f.

[5]

b) An 11 kV three-phase generator is connected to a 275 kV transmission line through an 11/275 kV transformer. The generator is star-connected with solidly grounded neutral. The 11 kV side of the transformer is delta-connected while the 275 kV side is star-connected with solidly grounded neutral. The sequence reactances of the generator and transformer are given in Table 5.1 on a common base.

	Rated voltage	Positive seq.	Negative seq.	Zero seq.
Generators	11 kV	$X''_{G1} = 0.05$ p.u.	$X''_{G2} = 0.05$ p.u.	$X''_{G0} = 0.01$ p.u.
Transformer	11 kV/275 kV	$X_{T1} = 0.1$ p.u.	$X_{T2} = 0.1$ p.u.	$X_{T0} = 0.1$ p.u.

Table 5.1: Sequence reactance of the generator and transformer

Calculate the magnitude of the fault current (in p.u.) for a solid line to ground (LG) fault at:

i) Line side of the transformer.

[3]

ii) Terminal of the generator.

[2]

c) If the 11 kV side of the transformer was star connected with solidly grounded neutral, calculate the magnitude of the fault current (in p.u.) for a solid line to ground (LG) fault at:

i) Line side of the transformer.

[2]

ii) Terminal of the generator.

[1]

iii) How would the line to line (LL) fault current change if the 11 kV side of the transformer is changed from delta (part b) to star with solidly grounded neutral (part c)?

[2]

Question 6

a) For a generator with a sinusoidal power angle characteristic, steady-state operation beyond a power angle of 90 could lead to instability. Justify this statement in the following ways:

i) With a physical explanation considering a small change in mechanical power input to the generator.

[4]

ii) Analytically using the characteristic equation of the system.

[4]

b) Two geographical areas of a 275 kV, 50 Hz system are connected by a transmission line with reactance $X = 0.3$ p.u. The voltage at the two ends of the line is tightly regulated at 1.0 p.u. The line is equipped with series compensation to increase the power transfer level. Making the necessary assumptions, calculate the following:

i) Critical clearing angle (in degrees) for 2.0 p.u. active power flow through the line and zero series compensation level considering a temporary self-clearing fault and zero power transfer between the two areas during the fault.

[4]

ii) Series compensation level (k in%) required to increase the maximum power transfer level by 50% compared to zero series compensation level.

[2]

iii) Critical clearing angle (in degrees) for 2.0 p.u. active power flow through the line and the same series compensation level as in part (ii) considering a temporary self-clearing fault and zero power transfer between the two areas during the fault.

[2]

iv) Active power flow (in p.u.) that can be allowed through the line in pre-fault condition if the critical clearing angle could be the same as in part (i).

[2]

c) The inertia constant of a generator in a nuclear power plant is 6 seconds. What is the physical significance of this statement?

[2]